

Full-Stack Developer Assignment – Mini-App

Deadline: 28th September, 2025

Objective

Build a **mini financial trading app** that demonstrates your ability to design secure APIs, integrate a frontend with the backend, and manage simple portfolio logic and make a 3-5 Minute video explaining your project.

Scope of Work

You will create a simplified version of a Financial App with the following features:

1. Authentication & KYC (Frontend + Backend)

- Users can **sign up and log in**.
- During sign-up, users must complete a **basic KYC form** (name, email, PAN number, upload dummy ID image).
- Store data securely in a database. File upload can be simulated (saving file locally or to cloud bucket).

2. Product Listing & Details (Frontend + Backend)

- Display a list of **dummy investment products** (e.g., 3–5 stocks or mutual funds seeded in DB).
- Each product should have: name, category, price per unit, and one key metric (like P/E ratio).
- A **Product Detail Page** should show more info (dummy data + chart using any charting library).

3. Transaction & Portfolio (Frontend + Backend)

- A logged-in user should be able to **"buy" a product** (enter units, confirm purchase).
 - On purchase:
 - Deduct balance from a virtual wallet (seed wallet with ₹100,000).
 - Store transaction in DB.
 - User's **Portfolio Dashboard** should show:
 - Total invested amount
 - Current value (use dummy calculation like units * price)
 - Returns (current value – invested)
 - A watchlist feature (user can add/remove products to watchlist)
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Technical Requirements

- **Backend:** Node.js + Express (or Nest.js) with PostgreSQL (or MongoDB if preferred). REST API endpoints for auth, KYC, products, transactions, portfolio.
- **Frontend:** React (or Next.js) with a clean UI.
- **Security:** Use JWT for authentication.
- **Extras (Optional for Bonus):**
 - Use Redis caching for product listing.
 - Add role-based login (Admin can see all users & transactions).

Deliverables

- GitHub repository with clear **README**:
 - Setup instructions
 - API documentation
 - **MUST : Simple Demo video (3 –5 mins screen recording walking through features).**
- Deployment optional (bonus points if deployed to Vercel/Netlify + Render/Railway).

Timeframe

You have **3 days** to complete this assignment, submit the assignment via the Google Form link along with your video as a drive link by **28th September (Sunday), 11:59 pm**. Please focus on **clean code, working features, and clarity of thought** rather than covering everything exhaustively.

Assignment Submit Link: <https://forms.gle/VT7uFim5B2Ao9KRB7>

